The empirical recurrence for OEIS A185849, proved by transfer matrix

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Abstract

OEIS A185849 counts arrays over $\{0, \dots, 2\}$ under a condition comparing a statistic of every 2×2 subblock with those of its neighbours, and carries an empirical recurrence of order 36 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical. The count is a walk count on $S = 630$ states and is C-finite; whether the conjectured recurrence annihilates it is then settled by a finite exact computation.

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1 The conjecture

OEIS A185849 is "Number of (n+1)X3 0..2 arrays with no 2X2 subblock diagonal sum less antidiagonal sum equal to any horizontal or vertical neighbor 2X2 subblock diagonal sum less antidiagonal sum.". It has offset 1 and begins

$$630, 11610, 215880, 4004914, 74358926, 1380412312, 25625942242, 475728999902, \dots$$

The entry states, as a conjecture contributed by its author and never marked settled:

> Empirical: a(n)=8*a(n-1)+106*a(n-2)+1154*a(n-3)+7341*a(n-4)+38354*a(n-5)+75897*a(n-6)+26908*a(n-7)+363811*a(n-8)-971504*a(n-9)-6880482*a(n-10)-1595117*a(n-11)+13890723*a(n-12)-50693559*a(n-13)-54673009*a(n-14)+185578667*a(n-15)-214035759*a(n-16)-910901251*a(n-17)+1570495381*a(n-18)+2337237818*a(n-19)-6890840995*a(n-20)-7626185862*a(n-21)+18279944018*a(n-22)+20105596552*a(n-23)-48993202848*a(n-24)-29327197968*a(n-25)+67685629696*a(n-26)+13126132416*a(n-27)-29811833984*a(n-28)-9414272512*a(n-29)-22788287488*a(n-30)+12806834176*a(n-31)+16026865664*a(n-32)-6554771456*a(n-33)-1887731712*a(n-34)+627769344*a(n-35)+55836672*a(n-36)

As of the "Last modified" line on the live entry (Oct 03 2025 03:15:03, revision 9) this is still recorded as empirical, and nothing on the entry records it as proved.

2 What is being counted

Write σ for its diagonal sum minus its antidiagonal sum, taken on a 2×2 subblock. The contiguous 2×2 subblocks of an $(n + 1) \times 3$ array over $\{0, \dots, 2\}$ form a grid of n rows and 2 columns, and σ labels that grid. The entry requires that no two subblocks that are neighbours horizontally or vertically carry the same value of σ.

3 The count is a walk count

The condition is on a PAIR of neighbouring subblocks, so nothing beyond the rows themselves has to be remembered. Take as state the 2-tuple of consecutive array rows whose subblock row has just been completed: the horizontal pairs inside that row decide which tuples are admissible at all, and the vertical pairs decide which tuple may follow which. A step appends one array row. An array of $n+1$ rows is a walk of $n-1$ steps and

$$a(n) \;=\; \iota^{\top} M^{\,n-1}\tau, \qquad \iota = \tau = (1,\dots,1)^{\top},$$

on the $S = 630$ admissible tuples.

The state does not compress. What a step needs of the past is the previous subblock row, and for $3 \geq 2$ that row already determines the 2 array rows it came from, so carrying the rows costs nothing and carrying anything smaller loses information. Of the $3^{2\cdot 3} = 729$ tuples of 2 rows, $S = 630$ survive the condition inside a single subblock row; the rest could not occur in a counted array at all.

Over the alphabet $\{0,\dots,2\}$ the statistic takes values in $[-4,4]$, so the grid it labels carries at most 9 different labels. At $n = 1$ the array has 2 rows and the grid is a single row of 2 subblocks, with only the neighbours inside that row; the entry's first term is 630, which the model reproduces along with every later term, and that is the cheapest check that the grid has been read with the right shape.

In particular a is C-finite. The alignment of the walk index with the entry's own n was checked against its published terms rather than assumed.

4 The proof

Lemma 1. *Let $q(t) = t^{36} - \sum_i c_i t^{36-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^{\top} M^{j} q(M)\tau$ for the corresponding walk index j, so the recurrence holds from some point on if and only if $\iota^{\top} M^{j} q(M)\tau = 0$ for all large j, and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+36} - \sum_i c_i M^{j+36-i}$. For the bound, M^S is an integer combination of $I, M, \dots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^{\top} M^j q(M)\tau$ obey the monic recurrence given by the characteristic polynomial of M; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. $\square$

Theorem 2.

$$a(n) \;=\; 8\,a(n-1)+106\,a(n-2)+1154\,a(n-3)+7341\,a(n-4)+38354\,a(n-5)+75897\,a(n-6)+26908\,a(n-7)+3638$$

for every $n > 36$.

Proof. The vector $w = q(M)\tau$ was formed by 36 matrix–vector products in exact integer arithmetic and $\iota^{\top} M^j w$ evaluated for $j = 0, 1, \dots,$ again exactly; those integers vanish from the index corresponding to $n = 36 + 1$ onwards and the run continued until the working vector was identically zero, which settles every larger j at once. $\square$

5 Verification

Three checks, each independent of the algebra above.

First, the model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry: the digraph is built by reading the entry's English, and a misreading gives different counts.

Second, the conjectured recurrence was evaluated directly on those published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Third, the annihilation test of Lemma 1 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

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[3] J. E. Hopcroft, R. Motwani and J. D. Ullman, *Introduction to Automata Theory, Languages, and Computation*, 3rd ed., Addison–Wesley, 2006.

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